

PROVISIONAL PATENT APPLICATION

INVENTOR DETAILS

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TITLE OF THE INVENTION

Emergency Dial-Code Based Safety Alert System with Live Location Sharing for Mobile Devices

BACKGROUND OF THE INVENTION

Personal safety, especially for women, children, elderly, and vulnerable individuals, has emerged as a critical issue in the modern digital era. While various mobile safety applications exist, they present several limitations:

1. **Internet Dependency** – Most existing SOS applications require active internet access to transmit alerts or location. In low-connectivity areas, these apps fail.
2. **Pre-Installed Applications** – Many solutions demand installation and configuration of third-party apps, which limits adoption.
3. **Balance/Recharge Requirement** – Conventional alerts through SMS or calls may fail if the user does not have sufficient balance.
4. **User Inaccessibility in Emergencies** – In stressful or dangerous situations, navigating through apps or multiple clicks is not feasible.
5. **Visibility Risk** – Most alert applications are visible on-screen, which may alert a potential attacker and escalate danger.

Thus, there exists a **gap** for a solution that:

- Works without internet or balance.
- Can be triggered discreetly with minimal effort.
- Shares continuous location until the emergency ends.
- Operates in stealth mode to protect the victim.

SUMMARY OF THE INVENTION

The present invention provides a **telecommunication-integrated mobile safety system** that enables users to discreetly trigger an emergency alert by dialing a **unique emergency code** (e.g., *999#) from the device's keypad. The system is designed to remain **active and tamper-resistant**, continuing to function even if an attacker attempts to switch off the mobile device, ensuring **uninterrupted emergency communication** until deactivated by the rightful user.

Once activated, the system performs the following operations:

1. **Stealth Activation** – The emergency protocol runs in the background without visible notifications, maintaining user discretion. A dummy “call failed” message may be displayed to mislead the attacker.
2. **Location Acquisition** – The GPS module silently retrieves real-time coordinates, re-enabling itself if disabled.
3. **Secure Message Dispatch** – The system sends the current location to predefined emergency contacts using **USSD or SMS protocols**, which function without internet or mobile balance.
4. **Continuous Tracking** – The system operates in a loop, transmitting updated location data at fixed intervals (e.g., every 30–60 seconds) to ensure real-time tracking.
5. **Tamper-Resistance & Shutdown Prevention** –
 - Once the system is triggered, it gains **priority execution rights** within the operating system.
 - The device’s **power button and shutdown functionality** are overridden or delayed until emergency dispatch is complete.
 - Even if the attacker long-presses the power button, the system continues transmitting location data in the background until a secure deactivation code (e.g., #999#) or PIN/password is entered.
 - This ensures that the emergency alert cannot be terminated by forceful shutdown.
6. **Fallback & Redundancy** – If USSD transmission fails, SMS is automatically attempted. Multiple layers of fail-safes ensure successful delivery.
7. **Secure Deactivation** – Only the authorized user can deactivate the system using a **hidden dial code or PIN entry**. All temporary logs are erased after deactivation.

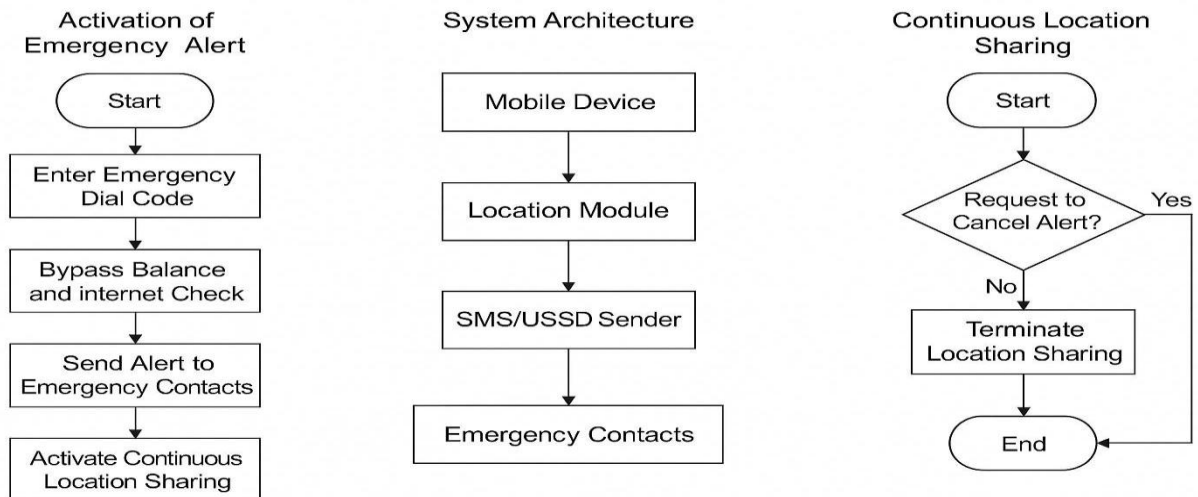
Novelty & Advantages

- Works without internet or balance.
- Activated through dial codes, not apps.
- Operates in stealth mode, invisible to attackers.
- **Tamper-proof: cannot be shut down by an attacker.**
- Provides continuous, real-time location tracking.

By incorporating **shutdown prevention and tamper-proof operation**, the invention overcomes a major limitation of conventional mobile safety apps, ensuring **uninterrupted emergency alerts** even under hostile conditions.

BRIEF DESCRIPTION OF THE DRAWINGS

Emergency Dial-Code Based Safety Alert System for Mobile Devices



Continuous Location Sharing and Deactivation

1. Start Emergency Alert System

- Triggered when the user dials the emergency code (e.g., *999#).
- The system activates in stealth mode, meaning it doesn't show anything visibly on the screen.

2. Acquire GPS Location

- The system fetches the current GPS coordinates of the device.
- If location is off, it prompts the GPS module silently to turn on.

3. Send Location to Emergency Contacts

- Sends an SMS or USSD-based message to pre-defined emergency numbers (parents, police).
- Message contains a Google Maps link or raw coordinates.

4. Wait for Interval (e.g., 1 Minute)

- The system sets a timer (e.g., 60 seconds) before the next update.
- This allows for continuous live tracking without draining battery instantly.

5. Recheck for Deactivation Code

- The system checks if the user has entered the deactivation code (e.g., #999#) or a secure PIN.
- This can be typed from the dialer again or accessed through a hidden input field.

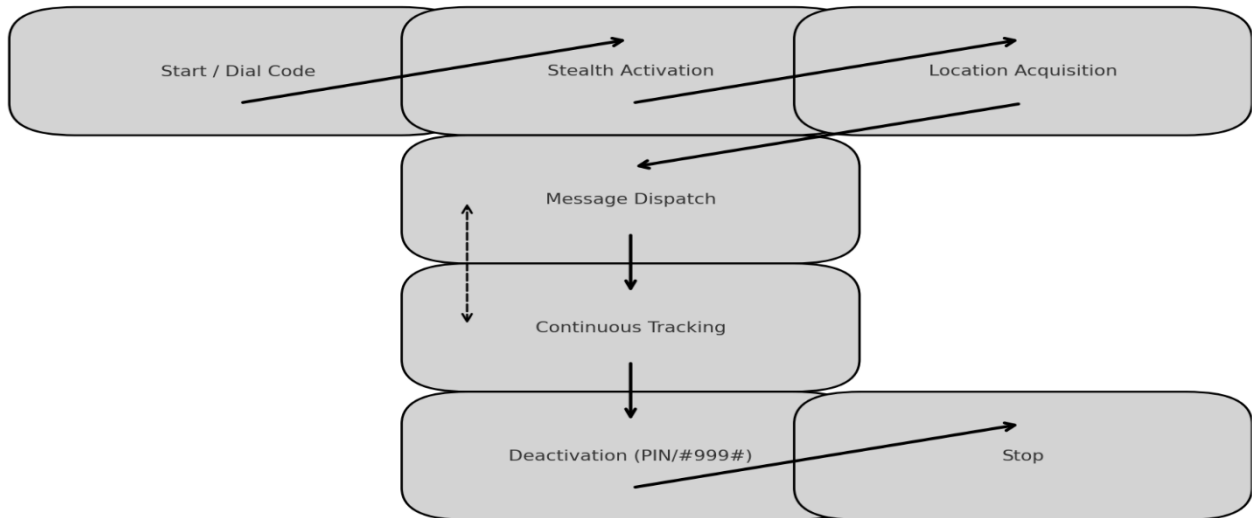
6. If Not Deactivated → Loop

- If the user hasn't stopped the alert, it loops back to fetch location and repeats the sending process.

7. If Deactivated → Stop Sharing

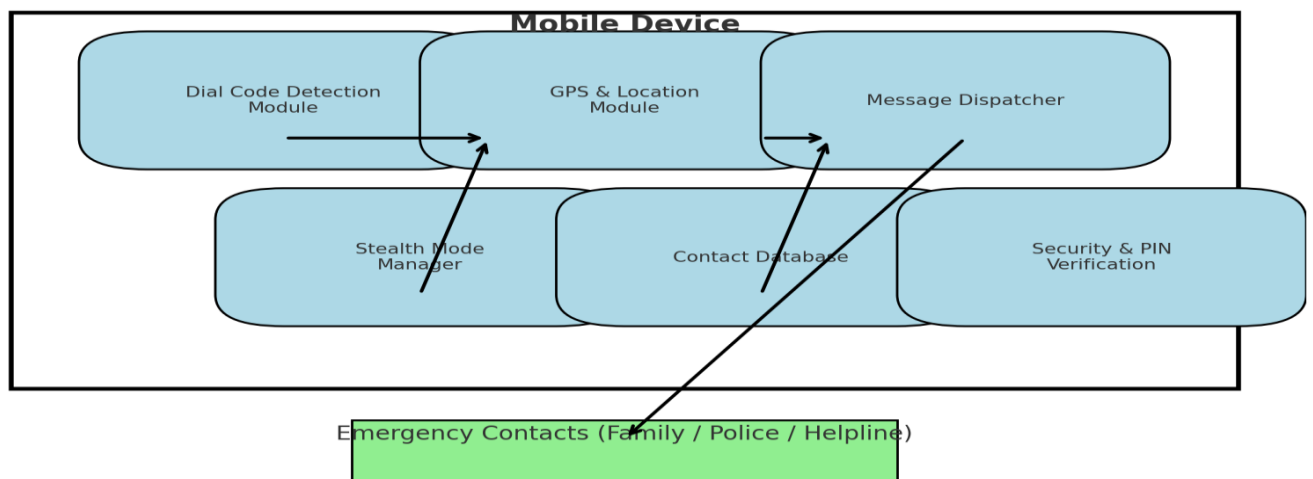
- All tracking stops immediately.
- The system deletes temporary logs, stops location services, and exits stealth mode.

Figure 1 - Flowchart of Emergency Process



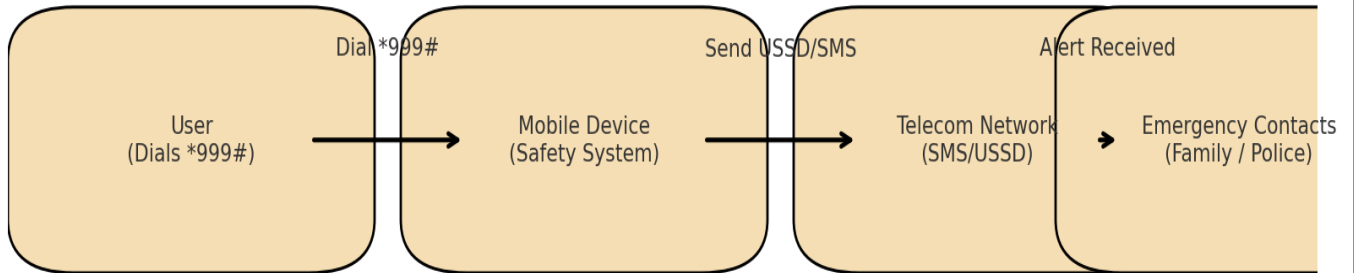
It shows the sequence: **Dial Code → Stealth Activation → Location Acquisition → Message Dispatch → Continuous Tracking → Deactivation → Stop**, with a feedback loop for continuous updates.

Figure 2 - System Architecture Diagram



It shows how the **Dial Code Detection Module, GPS, Message Dispatcher, Stealth Manager, Contact Database, and Security Layer** work together inside the mobile device, connecting to **Emergency Contacts** (family/police/helpline).

Figure 3 - User Interaction Diagram



***User dials 999# → Mobile Device activates safety system → Telecom Network transmits SMS/USSD → Emergency Contacts receive alerts.**

DETAILED DESCRIPTION OF THE INVENTION

The invention comprises a mobile-device-based safety alert system activated through a designated dial code. The detailed working of the system is explained below:

1. Emergency Activation via Dial Code

- The user dials a unique emergency code (e.g., *999#).
- The Dial Code Detection Module intercepts this input before it is routed as a normal call.
- The system interprets this code as an emergency trigger and initiates the alert process.
- A dummy “invalid code” or “call failed” message may be displayed to avoid suspicion.

2. Stealth Mode Operation

- Once activated, the system runs silently in the background.
- No sound, vibration, or notification is shown on the screen.
- This ensures the attacker remains unaware of the emergency alert.

3. Location Acquisition

- The GPS and Location Module fetches the device’s real-time latitude and longitude.
- If GPS is turned off, the module silently reactivates it.
- Location is formatted either as raw coordinates or as a Google Maps link.

4. Message Dispatch

- The Message Dispatcher sends the acquired location to preconfigured emergency contacts.
- Transmission occurs through SMS or USSD protocols, both of which work without internet.
- If one protocol fails, the system automatically switches to the other to ensure redundancy.
- The emergency message includes:
 - Alert header (e.g., 🚨 EMERGENCY ALERT 🚨)
 - User identity
 - Live location link
 - Timestamp
 - Disclaimer indicating automated SOS generation.

5. Continuous Live Tracking

- After the first alert, the system sets a timer (e.g., 30–60 seconds).
- Once the timer expires, the location is refreshed and retransmitted.
- This loop continues until the user deactivates the system.

6. Deactivation and Security

- The system can only be stopped using a secure deactivation code (e.g., #999#) or a PIN/password.
- Incorrect codes do not terminate the alert, preventing unauthorized deactivation.
- On successful deactivation, the system:
 - Stops all transmissions.
 - Deletes temporary logs.
 - Exits stealth mode.

7. Tamper Resistance & Shutdown Prevention

- The system overrides or delays the phone's power-off functionality.
- Even if an attacker attempts shutdown, the system ensures at least one emergency message is sent.
- Background processes remain active until the dispatch cycle is completed.

8. Contact Database

- The system includes a secure database of pre-registered emergency numbers.
- Numbers can include family, friends, police stations, or helpline services.

- The database is encrypted to prevent misuse and supports multiple entries for reliability.

9. **Stealth Functionality**

- No notifications, alerts, or pop-ups are shown during operation.
- Optional decoy screens (e.g., “Invalid Code”) may be displayed to avoid suspicion.

10. **Integration Possibilities**

- The system can be embedded in mobile operating systems as a built-in feature.
- It can be deployed as a telecom-integrated service in partnership with network providers.
- It may also be pre-installed as a system-level application.
- Integration with government emergency helplines or law enforcement systems enables direct response and monitoring.

11. **Key Technical Advantages**

- Works without internet or mobile balance.
- Provides continuous live location tracking.
- Fully operates in stealth mode.
- Tamper-resistant and cannot be disabled by attackers.
- Ensures redundancy through SMS and USSD protocols.
- Enhances personal safety for women, children, senior citizens, travelers, and field workers.

NOVEL ASPECTS OF THE INVENTION

1. **Dial-Code Based Activation** – Unlike existing safety apps that require opening and navigating applications, this invention is triggered by a simple dial code (e.g., *999#), enabling quick and discreet activation.
2. **Independence from Internet and Balance** – The system functions using USSD/SMS protocols, making it operable without mobile data or recharge balance, ensuring reliability in remote and low-resource environments.
3. **Stealth Mode Functionality** – The emergency process runs invisibly in the background, showing no visible notifications, sounds, or vibrations, and optionally displaying a decoy “call failed/invalid code” screen to avoid suspicion by an attacker.
4. **Continuous Live Location Sharing** – Unlike traditional SOS solutions that send a one-time alert, this invention enables ongoing location updates at fixed intervals (e.g., every 30–60 seconds) until deactivation, allowing real-time tracking.
5. **Secure Deactivation** – The system can only be stopped using a unique dial code or a secret PIN/password, preventing unauthorized termination of the emergency alert.

6. **Tamper Resistance & Shutdown Prevention** – A unique feature where the system overrides or delays the mobile device's shutdown process, ensuring at least one emergency message is transmitted even if an attacker tries to switch off the phone.
7. **Fallback & Redundancy Mechanism** – If the primary transmission method (USSD) fails, the system automatically switches to SMS, and vice versa, ensuring high delivery success rates.
8. **Encrypted Emergency Contact Database** – The system securely stores multiple pre-registered emergency numbers (family, police, helplines), protected through encryption to prevent misuse.
9. **Telecom-Level Integration** – Operates at the telecom protocol layer instead of depending solely on mobile applications, making it more resilient, faster, and less resource-dependent.
10. **Scalability for Public Safety** – Beyond personal use, the system can be integrated with law enforcement or government emergency networks for large-scale monitoring and rapid response.

ABSTRACT

The present invention relates to a mobile-device-based emergency safety system that enables users to discreetly trigger an alert by dialing a unique emergency code. Upon activation, the system operates in stealth mode, acquiring the user's GPS location and transmitting it to pre-registered emergency contacts via SMS or USSD protocols, thereby functioning without internet connectivity or mobile balance. Unlike conventional SOS applications, the system provides continuous live location updates at predefined intervals until securely deactivated through a separate dial code or PIN. The invention further incorporates tamper-resistant functionality that prevents premature termination by overriding or delaying the device's shutdown process, ensuring that at least one emergency alert is transmitted before power-off. The system also includes an encrypted contact database, redundant fallback mechanisms between SMS and USSD, and optional decoy displays to protect user discretion. By combining dial-code activation, balance-independent operation, stealth mode functionality, continuous tracking, and shutdown resistance, the invention offers a highly reliable and innovative solution for personal safety, particularly for women, children, senior citizens, and individuals in distress.